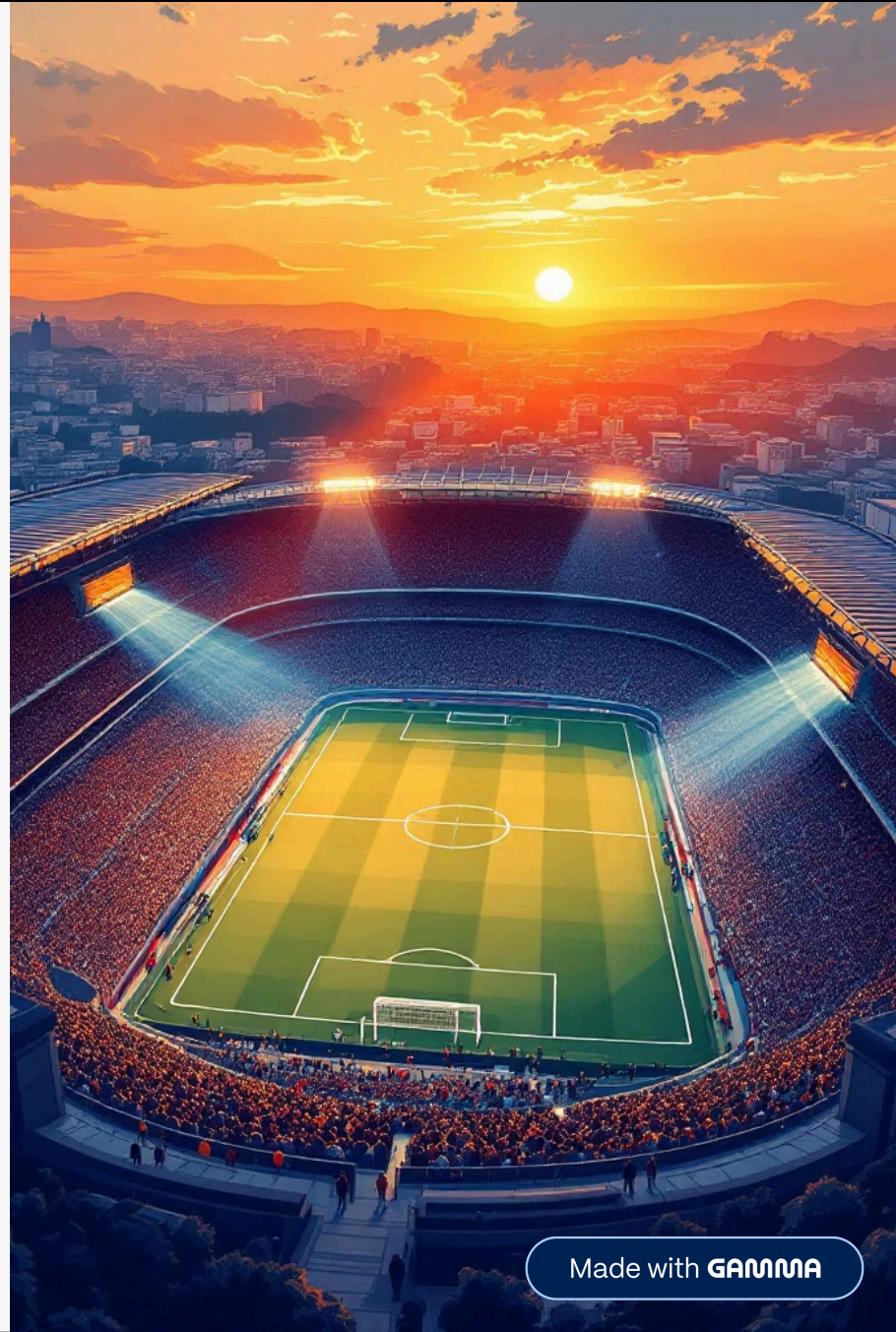
Soccer Match & Player Analytics – Project Overview

This project applies machine learning to rich European soccer data (2008–2016) to quantify player impact, surface team strengths, and predict match outcomes. Deliverables include performance metrics, predictive models, and visual dashboards to support coaching and analytical decisions.



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Dataset Snapshot

The dataset is comprehensive and structured for modeling:

- ~25,000 matches across 2008–2016
- 10,000 players from 11 European countries
- Player/team attributes (FIFA-style ratings)
- Match stats: goals, assists, fouls, possession, shots, cards
- Betting odds, formations, and contextual metadata

Rich temporal coverage and multiple feature types support classification, regression, and longitudinal analyses.

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Problem Statement – Why this matters

Teams collect vast volumes of match and player data but lack streamlined systems to convert raw feeds into actionable insights. Coaches need fast, reliable analytics to:

Evaluate player performance

Objective metrics for selection, substitution, and scouting.

Identify winning strategies

Link formations, possession profiles, and pressing to outcomes.

Predict match outcomes

Probabilistic forecasts for planning and tactical preparation.

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Project Objectives

Clear, prioritized goals that guide modeling, evaluation, and delivery:

01

Analyze performance

Aggregate per-match and season-level metrics (xG, defensive actions, involvement).

02

Feature discovery

Find key predictors (form, lineup continuity, possession transitions, odds).

03

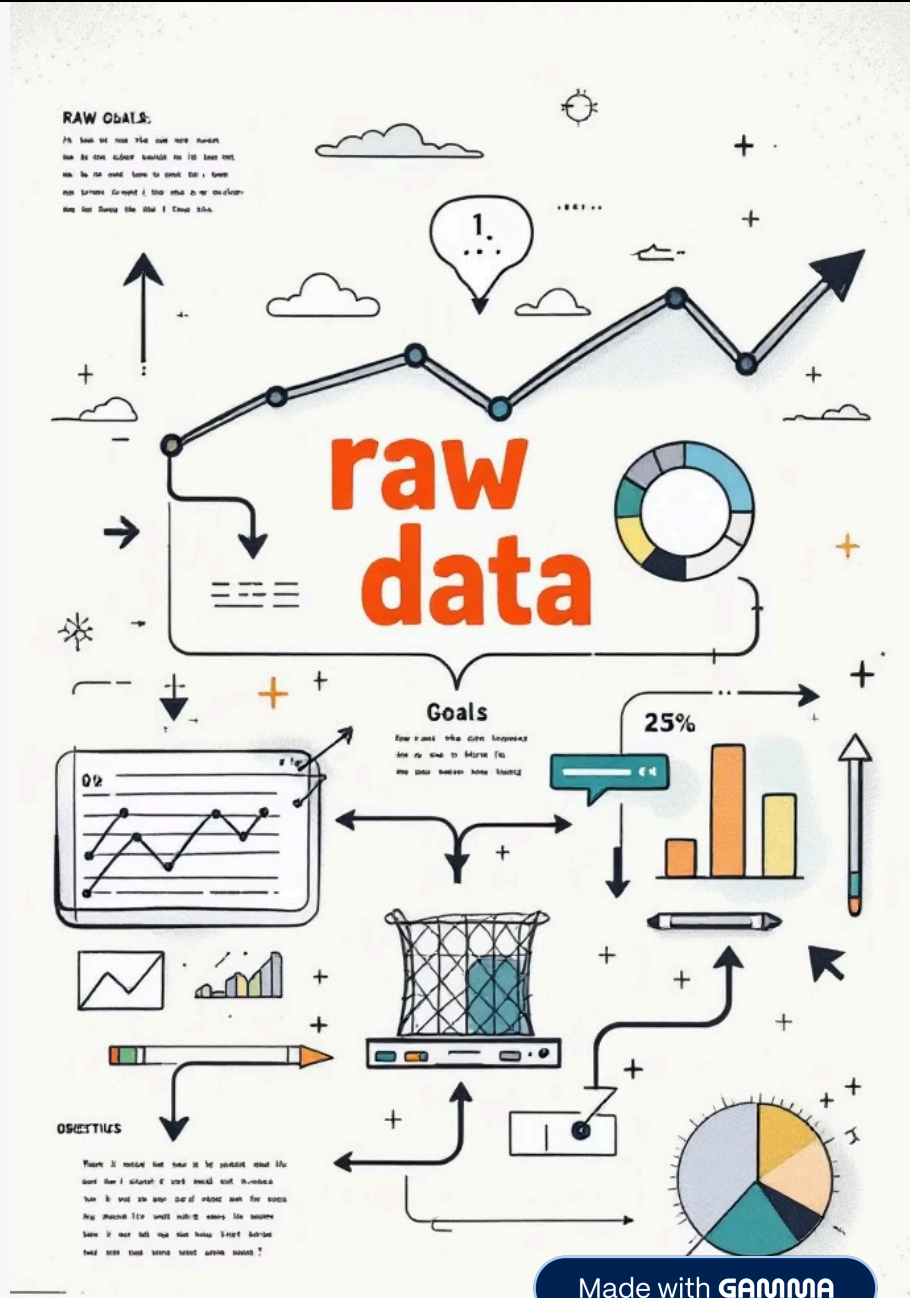
Predictive modeling

Train classifiers/regressors with cross-validation and calibrated probabilities.

04

Visualize & deploy

Interactive dashboards and reproducible pipelines for operational use.



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Problem Types & Modeling Approach

We will tackle three complementary analysis tracks:

- **Classification** — Predict win / draw / loss with probabilistic outputs; evaluate with log-loss and calibration.
- **Regression** — Predict goal counts and goal differential; use Poisson/negative binomial or gradient methods.
- **EDA** — Time-series form, lineup effects, home advantage, betting-odds calibration, and player impact analyses.

Focus on interpretability: feature importance, SHAP values, and simple baseline models for stakeholder trust.

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Tools, Technologies & Workflow

Python

Pandas, NumPy for data wrangling; pipeline orchestration in notebooks and scripts.

Statistical Analysis (ANOVA)

We also use statistical techniques like ANOVA to analyze differences between groups and identify important features

Visualization

Matplotlib/Seaborn and interactive dashboards (Plotly or Streamlit) for stakeholder consumption.

SQL & Data Ops

Relational joins, feature stores, and reproducible queries for consistent training sets.

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Real-World Impact & Next Steps

Practical uses and immediate steps to operationalize the analytics:

Coaching Decisions

Substitution timing, tactical adjustments, and opponent scouting informed by model outputs.

Performance Evaluation

Objective player valuations for selection, training focus, and talent ID.

Commercial Extensions

Calibrated prediction services for analytics firms or betting partners, with ethical guardrails.

Immediate next steps: finalize feature list, create training/validation splits, run baseline models, and prepare a prototype dashboard for a pilot season.

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